第8次作业总结

P475 题3

题目

Write a C++ function to delete the pair with key k from a hash table that uses linear probing. Show that simply setting the slot previously occupied by the deleted pair to empty does not solve the problem。 How must Get(Program 8.4) be modified so that a correct search is made in the situation when deletions are permitted? Where can a new key be inserted?

```
template<class K, class E> pair<K, E>* LinearProbing<K, E>::Get(const K& k)
{// Search the linear probing hash table ht (each bucket has exactly one slot)
 // for k, If a pair with this key is found, return a pointer to this pair; otherwise,
 //return 0.
    int i = h(k);        //home bucket
    int j;
    for(j = i; ht[j] && ht[j]->first !=k; ){
        j = (j + 1) % b;         //treat the table as circular
        if(j == i) return 0;         //back to start point
    }
    if(ht[j]->first == k) return ht[j];
    return 0;
}
// Program 8.4 线性探测法
```

参考答案

1. 因为get函数中for循环的循环条件是`ht[j] && hj[j]->first!=k`，若只是简单的将槽设置为空的话，for循环执行到该槽后将不再循环，如要搜寻的k值原本排在删除值的后面的话，则出现问题。
2. 可将被删除的槽点中的key设为一个不会出现的值

   假设不会出现的值为IMPOSSIBLE

```
template<class K,class E>
void Delete(const K& k)
{
    int i = h(K);
    int j;
    for (j = i; ht[j] && ht[j]->firtst!=k;)
    {
        j = (j + 1) & b;
        if (j == i) return 0;
    }
    if (ht[j]->first == k) ht[j] = IMPOSSIBLE;
}
```

3. Get函数中应将for循环的判断条件不变，依旧是`ht[j] && ht[j]!=k`

4. 新关键字可以插入到空桶与桶中关键字为IMPOSSIBLE的位置